Compensated Current Choice and Counterfactual Identification

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Abstract

An exact compensatory transfer for a finite current change puts that change on a common money scale; shrinking the change reveals the corresponding local slope element. Repeating the comparison across inherited states gives the direct comparative static when the target environment itself is observed locally. The problem here is the counterfactual case in which the target baseline choice is observed but the neighboring compensated comparisons needed for that direct calculation are not.

The observed comparisons form a directed graph. Circulations in its current projection eliminate arbitrary current-node values. Under cancellation, the transfers admit an additive representation, and the continuation differences identified by the data are exactly the image of the current cycle space under the continuation incidence map. Its dimension is

$$\operatorname{rank}\begin{pmatrix} D \\ Q \end{pmatrix} - \operatorname{rank} D.$$

One additional comparison contributes at most one continuation direction, and does so exactly when its current incidence is redundant while its joint current–continuation incidence is not. A primitive two-block design then identifies H_{xa} and H_{xx} although no sampled level $H(z)$ is point identified. More generally, shrinking circulations may annihilate every lower-order polynomial term while retaining a higher-order moment, so higher-order local functionals can be identified without lower-order jet recovery. These identified derivatives enter the target first-order condition and identify the counterfactual comparative static. Extensions to several environments and to state dynamics use standard potential and dynamical-systems arguments after the comparison-based identification step.

1 Introduction

A current choice may alter both the present account and what follows from it. At action x and inherited state a, compare the current bundle with one containing Δ more of the action and $p\Delta$ less of a monetary numeraire. Let $p_\Delta(x, a)$ be the compensating rate at indifference. On a common money scale,

$$p_\Delta(x, a) = \frac{W(x + \Delta, a) - W(x, a)}{\Delta},$$

where W includes the continuation consequences of the current action. Hence the limiting compensating rate

$$\pi(x, a) := \lim_{\Delta \downarrow 0} p_\Delta(x, a) = W_x(x, a)$$

is Samuelson's local slope element expressed in the numeraire. Figure 1 records the comparison and its limit.

Let $C(x,a)$ be the known current resource cost. At an interior strict local optimum,

$$\pi(x,a) = C_x(x,a).$$

Repeated comparisons across inherited states trace a regular local choice branch $x = g(a)$ and therefore its direct comparative static. Figure 2 gives this benchmark.

A two-stage insurance contract gives a useful running example. The current consequence records present coverage and other current contract terms after known monetary amounts have been netted out; the continuation consequence is a state-contingent renewal contract. An elicitation menu may hold the current consequence fixed while varying the renewal clause and offsetting the difference with a current premium transfer. Thus the same current node can appear with more than one continuation label even when a particular insurer's target menu links current coverage to a single renewal rule. The theory below concerns the information in such observed compound comparisons; it does not require the target menu itself to contain every recombination.

Now let A be observed locally and let B be a target environment whose baseline choice is observed but whose neighboring compensated comparisons are absent. Write

$$H(x,a) := (V \circ \Gamma_A)(x,a) - (V \circ \Gamma_B)(x,a).$$

The target response depends on derivatives of H. A zero sum of physical increments does not remove an unrestricted current value. Exact removal requires balance at every current node. If D is the node-edge incidence matrix, an edge vector r with

$$Dr = 0$$

is a circulation; equivalently, it belongs to the graph's cycle space. Current-node values then telescope, while the continuation loading Qr need not vanish. Figure 3 shows the closed current projection and the non-closing continuation lift.

Under an observable cancellation condition, transfers admit an additive representation. The continuation information that survives current cancellation is

$$Q(\ker D).$$

Its dimension is the rank gained by adjoining continuation incidence to current incidence. An additional comparison contributes a new continuation direction exactly when its current incidence is already spanned by the existing comparisons but its joint current–continuation incidence is not. Thus the information content of the comparison support is attributed edge by edge.

The finite geometry has a local consequence stronger than level recovery. A primitive design formed from two four-edge cycles identifies $H_{xa}(z_0)$ and $H_{xx}(z_0)$ while every sampled value $H(z)$ remains unidentified. More generally, shrinking circulation weights may annihilate all polynomial terms below order m while retaining a nonzero mth moment. The corresponding mth-order differential functional is then identified although every nonzero lower-order jet functional remains unidentified. The counterfactual first-order condition converts the two second-order continuation corrections into the target comparative static.

The order of the argument follows revealed preference: observed comparisons and finite consistency restrictions precede numerical representation. Samuelson (1938, 1948) formulate consumer theory in those terms, and Samuelson (1950) describes locally revealed price ratios as slope elements of the preference field. Debreu (1954) separates a preference ordering from its numerical representation, while Afriat (1967) constructs utility from finite expenditure data. Under partial observability, Gorno (2019) studies identification of preferences and Chambers, Echenique, and Lambert (2021) study recovery from growing finite binary experiments. The object identified here is narrower: continuation functionals sufficient for a counterfactual comparative static.

Money-metric utility and compensated demand are studied by Fuchs-Seliger (1989, 1990); Bhattacharya (2015) gives nonparametric money-metric welfare results for discrete choice. Quasilinear rationalizability and integrability are characterized by cyclical restrictions in Rochet (1987); Brown and Calsamiglia (2007); Nocke and Schutz (2017), and Castillo and Freer (2023) use a directed graph to construct a quasilinear rationalization. Makowski and Ostroy (2013) formulate preference revelation through linear-space duality and cyclic monotonicity. The graph relations below have a different role: circulations eliminate unrestricted current-node values and identify continuation differences. The additive representation belongs to the broader theory of cancellation and additive measurement (Kraft, Pratt, and Seidenberg, 1959; Scott, 1964; Doignon and Falmagne, 1974; Fishburn, Marcus-Roberts, and Roberts, 1988; Fishburn, 2001).

The maintained monetary structure is a transferable numeraire and a unique exact compensatory transfer for each observed binary comparison. Extensions after the main identification results use standard graph potentials, iterated-map terminology, and recursive methods (Elaydi, 2005; Kuznetsov, 2004; Stokey, Lucas, and Prescott, 1989).

2 The primitive compensated binary experiment

Let x denote a scalar current action, a an inherited state, and m a monetary numeraire. For each (x, a), let $W(x, a)$ be the money-metric value of the non-numeraire consequences of choosing x in state a. The current alternative (x, m) therefore has value

$$m + W(x, a). \tag{1}$$

All continuation consequences induced by the current action are included in W.

Fix (x, a) and $\Delta > 0$. Compare

$$B_0(x) = (x, m) \tag{2}$$

with

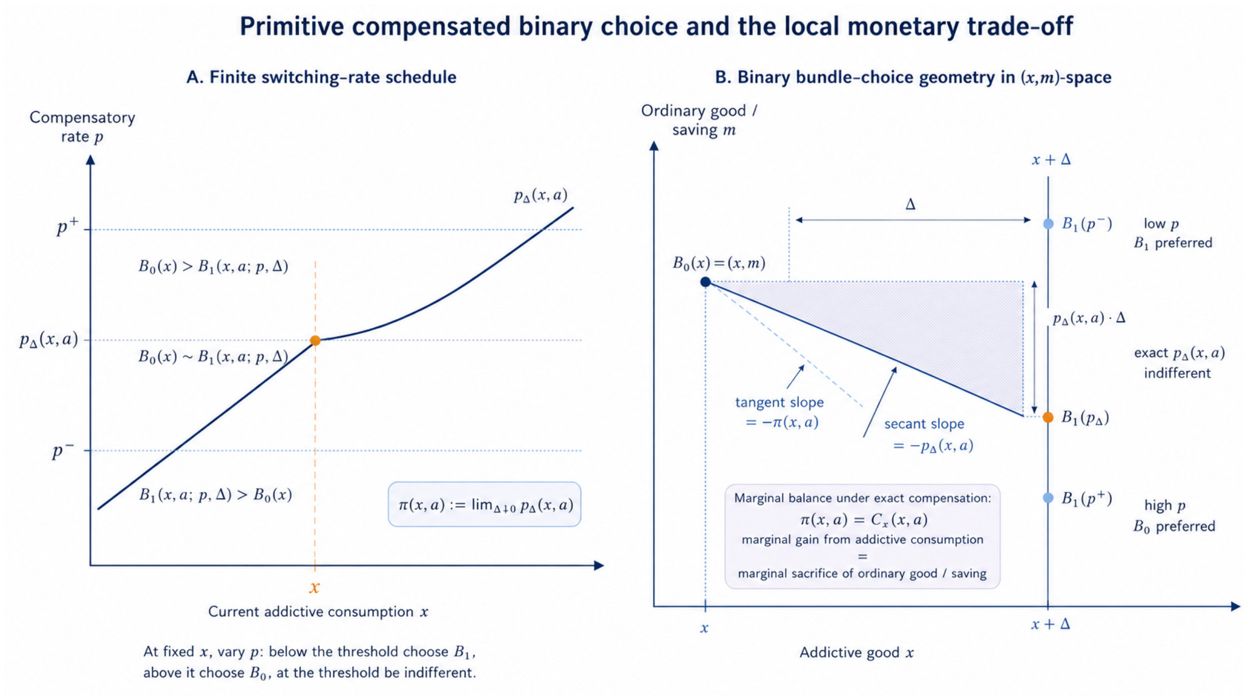
$$B_1(x, a; p, \Delta) = (x + \Delta, m - p\Delta), \tag{3}$$

where the continuation consequence is that induced by $(x + \Delta, a)$. All other current attributes are fixed. Only p is varied.

More generally, write a compound alternative as

$$(\xi, y, m),$$

where ξ is its current consequence and y its continuation consequence. For any pair belonging to the observer's comparison domain, the exact compensatory transfer is defined by varying money on one arm. In the insurance example, y is the renewal clause and the monetary adjustment is a premium transfer. The neighboring-action comparison above is the case in which the current action also determines its continuation consequence.



For each Δ, let $p_\Delta(x,a)$ be the compensating rate defined by

$$B_1(x,a;p_\Delta(x,a),\Delta) \sim B_0(x), \tag{4}$$

and let

$$\tau_\Delta(x,a) := p_\Delta(x,a)\Delta$$

be the exact compensatory transfer. With the orientation of Figure 1,

$$p < p_\Delta(x,a) \quad\Longrightarrow\quad B_1(x,a;p,\Delta) \succ B_0(x), \tag{5}$$

$$p = p_\Delta(x,a) \quad\Longrightarrow\quad B_1(x,a;p,\Delta) \sim B_0(x), \tag{6}$$

$$p > p_\Delta(x,a) \quad\Longrightarrow\quad B_1(x,a;p,\Delta) \prec B_0(x). \tag{7}$$

At the compensating rate,

$$m + W(x,a) = m - p_\Delta(x,a)\Delta + W(x+\Delta,a), \tag{8}$$

and therefore

$$\boxed{p_\Delta(x,a) = \frac{W(x+\Delta,a) - W(x,a)}{\Delta}.} \tag{9}$$

Thus $p_\Delta(x,a)$ is the compensating variation per unit of the finite current increment.

Following Samuelson's slope-element terminology, define the limiting compensating rate

$$\boxed{\pi(x,a) := \lim_{\Delta\downarrow 0} p_\Delta(x,a).} \tag{10}$$

If $W(\cdot,a)$ is differentiable at x, equation (9) is its right difference quotient and therefore

$$\boxed{W_x(x,a) = \pi(x,a).} \tag{11}$$

If π is continuously differentiable near (x,a), then

$$W_{xa} = \pi_a, \qquad W_{xx} = \pi_x. \tag{12}$$

3 The local choice function

Let the current choice problem have net monetary value $W(x,a) - C(x,a)$, where $C(x,a)$ is the resource cost of action x in inherited state a. An interior optimum satisfies

$$W_x(x,a) = C_x(x,a). \tag{13}$$

By (11), this condition is observable wherever the compensated experiment can be conducted:

$$\boxed{\pi(x,a) = C_x(x,a).} \tag{14}$$

Set

$$F(x,a) := \pi(x,a) - C_x(x,a). \tag{15}$$

The regular zero set of F defines the local choice function.

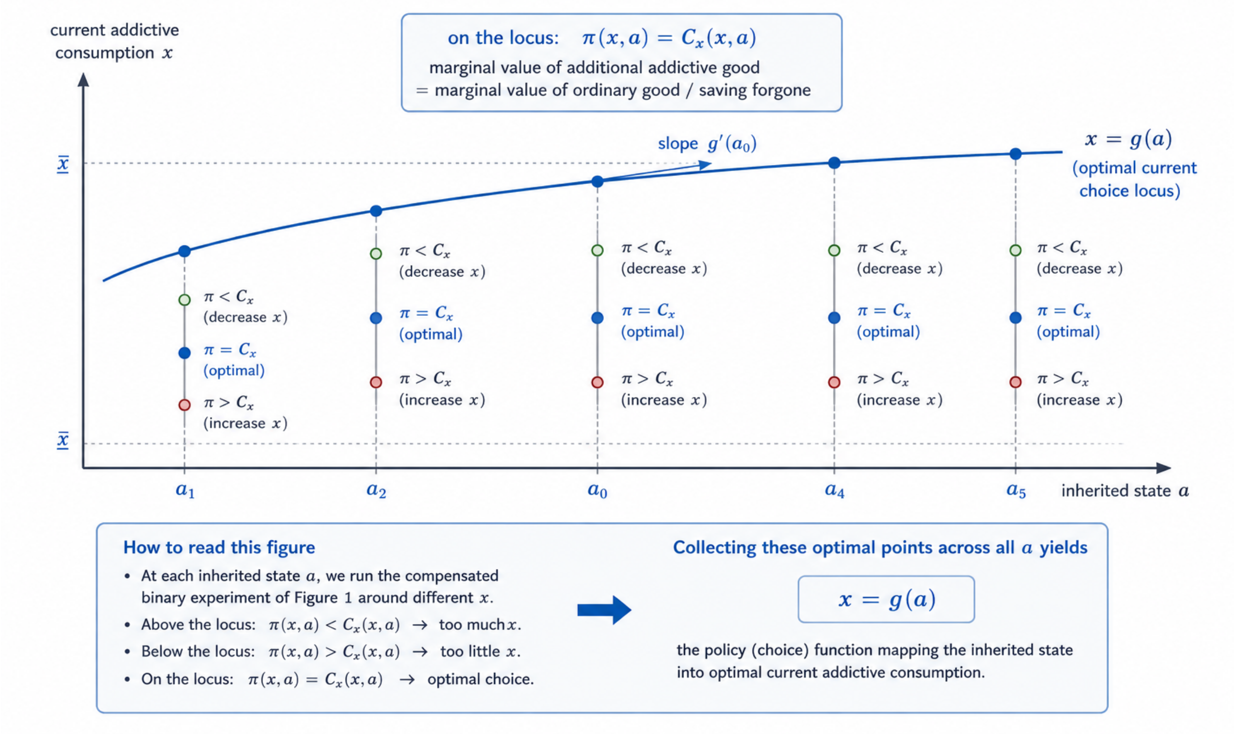


Figure 2. From local compensated choices to the optimal current choice locus $x = g(a)$

Repeating the compensated binary experiment of Figure 1 across inherited states

Suppose F is continuously differentiable near (x_0, a_0), $F(x_0, a_0) = 0$, and

$$C_{xx}(x_0,a_0) - \pi_x(x_0,a_0) > 0. \tag{16}$$

The implicit-function theorem gives a unique differentiable local choice branch $x = g(a)$ with

$$\boxed{F(g(a),a) = 0} \tag{17}$$

and

$$\boxed{g'(a) = \frac{\pi_a(g(a),a) - C_{xa}(g(a),a)}{C_{xx}(g(a),a) - \pi_x(g(a),a)}.} \tag{18}$$

When nearby compensated comparisons are observed, (18) is the direct comparative static.

3.1 The counterfactual problem

Let nearby compensated comparisons be observed in environment A. In environment B, suppose the baseline current choice $g_B(a_0)$ is observed but nearby compensated comparisons are not. Write

$$W_A = u_A + V \circ \Gamma_A, \qquad W_B = u_B + V \circ \Gamma_B.$$

This notation defines the structural counterfactual object of interest. No numerical values of V are used here to rationalize the finite comparison data. Section 4 instead characterizes when the observed compensatory transfers themselves admit an additive current–continuation representation. When the structural model is maintained, its continuation values give one such representation on the observed labels.

The compensated observations in A identify the local derivatives of W_A. The target response also depends on the derivatives of

$$H(x, a) := (V \circ \Gamma_A)(x, a) - (V \circ \Gamma_B)(x, a).$$

The problem is to identify finite and local functionals of H from the compensated comparisons that are observed.

4 Comparison graphs and cancellation

Consider a finite set E of observed compensated binary comparisons. Any current-side amount known in money units from the protocol is first netted from the raw compensatory transfer. τ_e denotes this adjusted exact transfer. Let

$$\mathcal{X} = \{\xi_0, \dots, \xi_N\}$$

be the distinct *current nodes* appearing in those comparisons after this known adjustment. A current node records the current-side attributes whose monetary value remains unrestricted. Thus two alternatives with the same current bundle may use the same node even when a known regime-specific current term differs; that known term has already been removed from τ_e. Each comparison e has a tail $s(e)$ and a head $t(e)$.

Let $D \in \mathbb{Z}^{(N+1)\times|E|}$ be the directed node-edge incidence matrix, whose eth column is

$$d_e = \mathbf{e}_{t(e)} - \mathbf{e}_{s(e)}. \tag{19}$$

Thus $Dz = 0$ means that the integer edge collection $z \in \mathbb{Z}^E$ enters and leaves every current node equally often. Thus $Dz = 0$ is exact node balance.

The protocol also records the signed continuation consequences carried by each comparison. After fixing a finite set of continuation labels and one normalization, write the continuation contrast of edge e as $q_e \in \mathbb{Z}^m$, and collect these columns in

$$Q \in \mathbb{Z}^{m\times|E|}.$$

Let $\tau = (\tau_e)_{e\in E} \in \mathbb{R}^E$.

Definition 4.1 (Cancellation)**.** The finite compensated data satisfy *cancellation* if

$$\boxed{Dz = 0, \quad Qz = 0 \quad \Longrightarrow \quad z^{\mathsf{T}}\tau = 0 \qquad (z \in \mathbb{Z}^E).} \tag{20}$$

Cancellation requires the monetary sum to vanish whenever both the current incidence and the continuation contrast vanish.

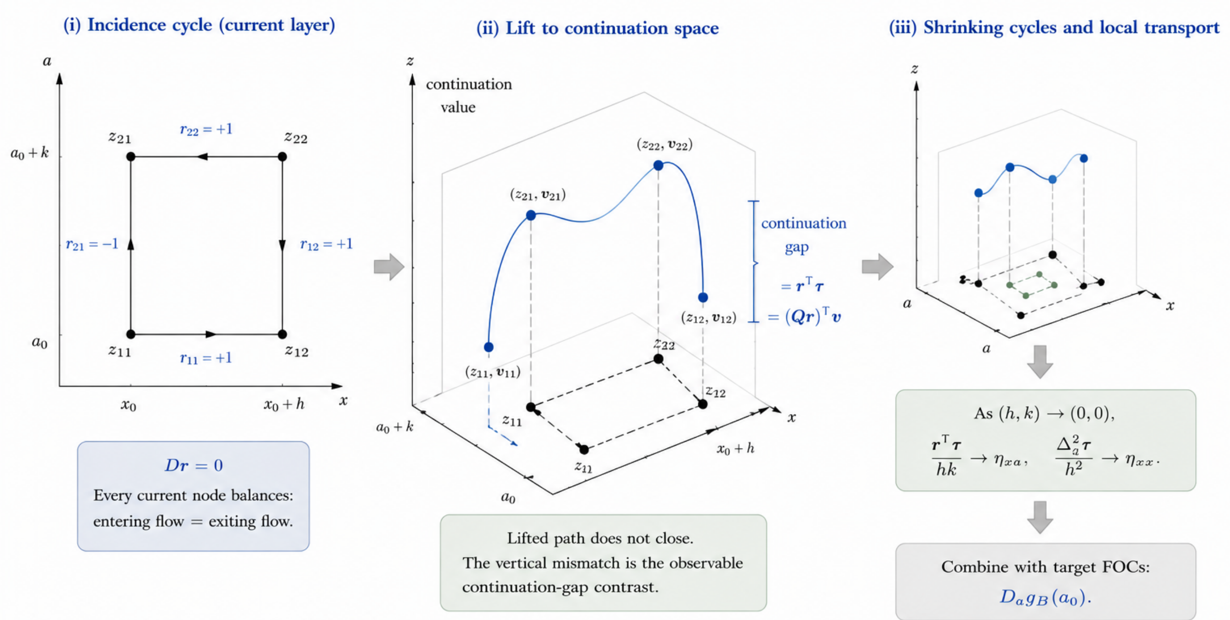
Theorem 4.2 (Additive monetary representation)**.** *Cancellation holds if and only if there exist a current-node potential $u \in \mathbb{R}^{N+1}$ and a continuation potential $v \in \mathbb{R}^{m}$ such that*

$$\boxed{\tau = D^{\mathsf{T}} u + Q^{\mathsf{T}} v.} \tag{21}$$

The representation is unique up to the null space of the observed incidence and continuation matrices. The potentials (u, v) are representation coordinates obtained from the finite data; they are not the structural functions (u_J, V) introduced in Section 3.

Figure 3 illustrates a node-incidence cycle and its continuation lift.

Figure 3. From Incidence Cycles to Local Derivative Transport



Notes: Panel (*i*) shows a primitive incidence cycle with incidence balance $Dr = 0$. Panel (*ii*) lifts the same base points into continuation space; the lifted path is generally non-closed. The signed vertical mismatch equals the observable continuation-gap contrast $r^{\mathsf{T}}\tau = (Qr)^{\mathsf{T}} v$. Panel (*iii*) uses shrinking cycles to deliver the local mixed and pure second derivatives (η_{xa}, η_{xx}), which, together with target FOCs, identify the comparative-static response $D_a g_B(a_0)$.

Corollary 4.3 (Circulation equation)**.** *Under cancellation, every current circulation $z \in \mathbb{Z}^E$ with $Dz = 0$ satisfies*

$$\boxed{z^{\mathsf{T}}\tau = (Qz)^{\mathsf{T}} v.} \tag{22}$$

$z^{\mathsf{T}} D^{\mathsf{T}} u = (Dz)^{\mathsf{T}} u = 0$, so the current-node values disappear from every circulation equation.

Theorem 4.4 (Finite identification)**.** *Suppose cancellation holds, and consider the model set*

$$\mathcal{M}(\tau) = \{(u, v) : D^{\mathsf{T}} u + Q^{\mathsf{T}} v = \tau\}.$$

For a target continuation contrast $c \in \mathbb{Z}^m$, the value $c^{\mathsf{T}} v$ is point identified on $\mathcal{M}(\tau)$ if and only if

$$\boxed{\exists\, k \geq 1,\ z \in \mathbb{Z}^E \quad \text{such that} \quad Dz = 0, \qquad Qz = kc.} \tag{23}$$

When (23) *holds,*

$$\boxed{c^\mathsf{T} v = \frac{z^\mathsf{T}\tau}{k}.} \tag{24}$$

Otherwise, relative to (21)*,*

$$\boxed{\{c^\mathsf{T} v : (u, v) \in \mathcal{M}(\tau)\} = \mathbb{R}.} \tag{25}$$

The identified continuation contrasts are exactly the image, after finite replication, of the integer circulation lattice under Q.

Unless stated otherwise, finite identification below is relative to the additive representation (21). Local identification is relative to the reduced-form C^m continuation functions satisfying the observed circulation equations. If a fixed common V and fixed continuation maps impose further cross-point restrictions, the corresponding structural identified set may be smaller.

5 Local identification from shrinking circulations

Let $z = (x, a)$ and fix $z_0 = (x_0, a_0)$. In a local design n, let

$$z_{1n}, \ldots, z_{M_n n}$$

be the sampled design locations and write

$$A(z) := (V \circ \Gamma_A)(z), \qquad B(z) := (V \circ \Gamma_B)(z), \qquad H(z) := A(z) - B(z).$$

Order the continuation labels in Q_n as those attached to

$$A(z_{1n}), \ldots, A(z_{M_n n}), B(z_{1n}), \ldots, B(z_{M_n n}).$$

When the finite representation is interpreted through the structural model, the continuation potential v assigns these labels the values $A(z_{in})$ and $B(z_{in})$, up to the common normalization. For sharp local statements we work directly with the reduced-form functions A and B satisfying the observed circulation equations. The primitive compensated comparisons in design n have observed current incidence D_n, continuation contrast Q_n, and exact transfers τ_n.

5.1 The target environment and the observed comparisons

The target environment is defined on a neighborhood of z_0 by

$$W_B(x, a) = u_B(x, a) + V(\Gamma_B(x, a)).$$

Let $\mathcal{E}_n$ denote the compensated comparisons observed in local design n. The family $\{\mathcal{E}_n\}$ need not contain every comparison formed from nearby alternatives in B. It may also contain elicitation menus that pair a current consequence ξ with a continuation contract $\Gamma_J(z)$ although that compound alternative is not part of the target choice set. In the insurance example, these are contract variants that preserve current coverage while substituting a different state-contingent renewal clause. Identification below is relative to $\{\mathcal{E}_n\}$; Γ_B remains defined on the full local choice set.

5.2 Continuation contrasts from the cycle space

Let

$$S_n := (I_{M_n}\ I_{M_n}), \qquad T_n := (I_{M_n}\ 0).$$

The integer circulation lattice relevant for an A–B contrast is

$$\mathcal{C}_n^{\mathbb{Z}} := \left\{ r \in \mathbb{Z}^{E_n} : D_n r = 0,\ S_n Q_n r = 0 \right\}. \tag{26}$$

These are finite replication and cancellation relations among the observed comparisons.

Their observed equations may be combined linearly. Define the real cycle space

$$\mathcal{C}_n := \operatorname{span}_{\mathbb{R}} \mathcal{C}_n^{\mathbb{Z}} = \left\{ r \in \mathbb{R}^{E_n} : D_n r = 0,\ S_n Q_n r = 0 \right\}. \tag{27}$$

The equality follows because D_n and Q_n have integer coefficients: rational kernel vectors span the real kernel, and denominators can be cleared to obtain integer circulations.

For $r \in \mathcal{C}_n$, the aggregate continuation contrast has the form

$$Q_n r = (\lambda, -\lambda), \qquad \lambda = T_n Q_n r.$$

The resulting contrast space is

$$\boxed{\Lambda_n := T_n Q_n \mathcal{C}_n \subseteq \mathbb{R}^{M_n}.} \tag{28}$$

Under cancellation, every $r \in \mathcal{C}_n^{\mathbb{Z}}$ satisfies

$$r^{\mathsf{T}} \tau_n = \sum_i \lambda_i H(z_{in}).$$

The same identity holds by linearity for every $r \in \mathcal{C}_n$:

$$r^{\mathsf{T}} \tau_n = \sum_i \lambda_i H(z_{in}), \qquad Q_n r = (\lambda, -\lambda). \tag{29}$$

5.3 Shrinking circulations and finite differences

Fix an open neighborhood U of z_0 containing the local design points. A smooth perturbation q is observationally null when

$$\sum_i \lambda_i q(z_{in}) = 0 \qquad \text{for every } n \text{ and every } \lambda \in \Lambda_n. \tag{30}$$

Identification over the reduced-form smooth class has the usual null-space form: a local linear functional is point identified exactly when it vanishes on every observationally null perturbation. This observation is used below only to state sharpness; the identifying formulas come directly from finite differences.

For $t = (t_0, t_x, t_a, t_{xx}, t_{xa}, t_{aa}) \in \mathbb{R}^6$, let

$$\mathcal{L}_t f := t_0 f + t_x f_x + t_a f_a + t_{xx} f_{xx} + t_{xa} f_{xa} + t_{aa} f_{aa} \quad \text{at } z_0. \tag{31}$$

Let P_n have ith row

$$\left(1, \delta x_{in}, \delta a_{in}, \frac{1}{2}\delta x_{in}^2, \delta x_{in}\delta a_{in}, \frac{1}{2}\delta a_{in}^2\right),$$

where $(\delta x_{in}, \delta a_{in}) = z_{in} - z_0$.

Theorem 5.1 (Shrinking finite differences)**.** *Suppose the support radius tends to zero. Let $\lambda_n \in \Lambda_n$ and $\alpha_n \neq 0$ satisfy*

$$\frac{P_n^{\mathsf{T}}\lambda_n}{\alpha_n} \longrightarrow t \tag{32}$$

and

$$\frac{1}{|\alpha_n|}\sum_i |\lambda_{in}|(\delta x_{in}^2 + \delta a_{in}^2) = O(1). \tag{33}$$

Then, for every C^2 function f near z_0,

$$\boxed{\frac{1}{\alpha_n}\sum_i \lambda_{in} f(z_{in}) \longrightarrow \mathcal{L}_t f(z_0).} \tag{34}$$

Under cancellation, the left side for $f = H$ is observed as a normalized circulation sum of compensatory transfers. Hence $\mathcal{L}_t H(z_0)$ is point identified.

Proof. Taylor expansion gives

$$f(z_{in}) = P_{n,i\cdot}\, j_{z_0}^2 f + R_{in}, \qquad R_{in} = o(\delta x_{in}^2 + \delta a_{in}^2)$$

uniformly over the shrinking support. Equations (32)–(33) give the limit; equation (29) makes it observable for H. $\square$

The familiar centered second difference and mixed rectangle are the cases

$$(1, -2, 1)/h^2 \quad \text{and} \quad (1, -1, -1, 1)/(hk),$$

provided the corresponding weights belong to Λ_n.

5.4 Second derivatives without level recovery

The comparison graph is the current projection of the observed compound alternatives. Let $\mathcal{A}_n$ be the set of compound alternatives observed in design n and write a typical element as (ξ, y), omitting money. Each observed comparison edge e has the form

$$(\xi_{s(e)}, y_e^-) \quad \text{versus} \quad (\xi_{t(e)}, y_e^+),$$

so its current incidence is

$$d_e = \mathbf{e}_{t(e)} - \mathbf{e}_{s(e)}$$

and its continuation contrast is

$$q_e = [y_e^+] - [y_e^-].$$

Proposition 5.2 (Dimension of the continuation image)**.** *Let D and Q be the current and continuation incidence matrices of any finite observed compound-menu support, and define*

$$\mathcal{T} := Q(\ker D).$$

Then

$$\boxed{\dim \mathcal{T} = \operatorname{rank}\begin{pmatrix} D \\ Q \end{pmatrix} - \operatorname{rank} D.} \tag{35}$$

Hence the observed support carries nonzero continuation information after current cancellation if and only if

$$\operatorname{rank}\begin{pmatrix} D \\ Q \end{pmatrix} > \operatorname{rank} D.$$

For the local A–B contrasts of Section 5, set

$$M_n := \begin{pmatrix} D_n \\ S_n Q_n \end{pmatrix}.$$

Then

$$\boxed{\dim \Lambda_n = \operatorname{rank}\begin{pmatrix} M_n \\ T_n Q_n \end{pmatrix} - \operatorname{rank} M_n.} \tag{36}$$

Thus the comparison support yields a nonzero A–B contrast exactly when adding the contrast incidence $T_n Q_n$ raises the rank after current incidence and common continuation incidence have been removed.

Proof. The restriction of Q to $\ker D$ has kernel $\ker D \cap \ker Q$. Rank–nullity gives

$$\dim Q(\ker D) = \dim \ker D - \dim(\ker D \cap \ker Q).$$

Since

$$\ker D \cap \ker Q = \ker \begin{pmatrix} D \\ Q \end{pmatrix},$$

(35) follows. Apply the same argument to the restriction of $T_n Q_n$ to $\ker M_n$ to obtain (36). □

Theorem 5.3 (Rank increment from one comparison)**.** *Let*

$$\nu(D, Q) := \operatorname{rank}\begin{pmatrix} D \\ Q \end{pmatrix} - \operatorname{rank} D = \dim Q(\ker D).$$

Add one observed comparison with current-incidence column d and continuation-incidence column q. If the comparison introduces a new current node or continuation label, embed the old matrices in the enlarged coordinate spaces by zero padding. Write

$$D^+ = (D \;\, d), \qquad Q^+ = (Q \;\, q).$$

Then

$$\boxed{\nu(D^+, Q^+) - \nu(D, Q) \in \{0, 1\}.} \tag{37}$$

The increment equals one if and only if

$$\boxed{d \in \operatorname{col} D \quad \text{and} \quad \begin{pmatrix} d \\ q \end{pmatrix} \notin \operatorname{col}\begin{pmatrix} D \\ Q \end{pmatrix}.} \tag{38}$$

Thus a comparison adds continuation information exactly when its current incidence is redundant but its joint current–continuation incidence is not.

For the local A–B contrast space, let

$$m = \begin{pmatrix} d \\ Sq \end{pmatrix}, \qquad g = Tq,$$

where S and T are the common- and contrast-incidence maps used in Section 5. With

$$M^+ = (M\ m), \qquad G^+ = (TQ\ g),$$

the contrast dimension $\dim \Lambda$ *likewise increases by one exactly when*

$$m \in \operatorname{col} M \quad \textit{and} \quad \begin{pmatrix} m \\ g \end{pmatrix} \notin \operatorname{col} \begin{pmatrix} M \\ TQ \end{pmatrix}. \tag{39}$$

Proof. Let

$$A = \begin{pmatrix} D \\ Q \end{pmatrix}, \qquad a = \begin{pmatrix} d \\ q \end{pmatrix}.$$

Adding one column changes each of $\operatorname{rank} A$ and $\operatorname{rank} D$ by either zero or one. If $d \notin \operatorname{col} D$, then $a \notin \operatorname{col} A$, so both ranks rise by one and their difference is unchanged. If $d \in \operatorname{col} D$, the rank of D is unchanged, while the rank of A rises by one precisely when $a \notin \operatorname{col} A$. This proves (37)–(38). The A–B statement is the same argument applied to the pair (M, TQ). □

Corollary 5.4 (Comparison lower bound)**.** *Let the current projection of a finite comparison support have V current nodes, E observed comparisons, and c connected components. Then*

$$\boxed{\dim Q(\ker D) \leq E - V + c.} \tag{40}$$

Consequently, any support carrying k linearly independent continuation directions after current cancellation must satisfy

$$\boxed{E \geq V - c + k.} \tag{41}$$

For a connected current graph, $E \geq V - 1 + k$.

Proof. For a node-edge incidence matrix,

$$\operatorname{rank} D = V - c.$$

Hence

$$\dim \ker D = E - V + c.$$

Since $Q(\ker D)$ is an image of $\ker D$, its dimension cannot exceed (40). Rearranging gives (41). □

Corollary 5.5 (Single-valued continuation support)**.** *Suppose that, on the observed compound alternatives, the continuation consequence is a function of the current node: there is a map ρ such that every observed alternative has the form $(\xi, \rho(\xi))$. Then*

$$Qr = 0 \qquad \textit{for every } r \textit{ with } Dr = 0.$$

Hence current circulations carry no nonzero continuation contrast.

Proof. Let R map the basis vector of current node ξ to the basis vector of the continuation label $\rho(\xi)$. For every observed edge,

$$q_e = R(\mathbf{e}_{t(e)} - \mathbf{e}_{s(e)}) = Rd_e.$$

Thus $Q = RD$, and $Dr = 0$ implies $Qr = 0$. □

A nonzero continuation image therefore requires the rank gain in Proposition 5.2. In particular, continuation cannot be single-valued in the current node on the relevant observed support. In the running insurance example, the required non-singleton fiber is simply the availability of more than one renewal clause for the same current contract consequence. If the future consequence is technologically fixed by the current consequence, the image is zero.

Let

$$\gamma(z) := [\Gamma_A(z)] - [\Gamma_B(z)], \qquad \gamma(z)^\mathsf{T} v = H(z).$$

For the mixed derivative, set

$$z_{00,n} = (x_0, a_0), \quad z_{10,n} = (x_0 + h_n, a_0), \quad z_{01,n} = (x_0, a_0 + k_n), \quad z_{11,n} = (x_0 + h_n, a_0 + k_n),$$

where $h_n, k_n \to 0$ and $h_n k_n \neq 0$.

Choose four current consequences

$$\xi^R_{1n}, \xi^R_{2n}, \xi^R_{3n}, \xi^R_{4n}$$

and observe the following four compensated menus, oriented from left to right:

$$(\xi^R_{1n}, \Gamma_B(z_{00,n})) \longrightarrow (\xi^R_{2n}, \Gamma_A(z_{00,n})), \tag{42}$$
$$(\xi^R_{2n}, \Gamma_A(z_{10,n})) \longrightarrow (\xi^R_{3n}, \Gamma_B(z_{10,n})), \tag{43}$$
$$(\xi^R_{3n}, \Gamma_B(z_{11,n})) \longrightarrow (\xi^R_{4n}, \Gamma_A(z_{11,n})), \tag{44}$$
$$(\xi^R_{4n}, \Gamma_A(z_{01,n})) \longrightarrow (\xi^R_{1n}, \Gamma_B(z_{01,n})). \tag{45}$$

Their continuation contrasts are

$$+\gamma(z_{00,n}), \quad -\gamma(z_{10,n}), \quad +\gamma(z_{11,n}), \quad -\gamma(z_{01,n}).$$

For action second derivative, let

$$z_{+,n} = (x_0 + h_n, a_0), \qquad z_0 = (x_0, a_0), \qquad z_{-,n} = (x_0 - h_n, a_0),$$

choose four further current consequences

$$\xi^X_{1n}, \xi^X_{2n}, \xi^X_{3n}, \xi^X_{4n},$$

and observe

$$(\xi^X_{1n}, \Gamma_B(z_{+,n})) \longrightarrow (\xi^X_{2n}, \Gamma_A(z_{+,n})), \tag{46}$$
$$(\xi^X_{2n}, \Gamma_A(z_0)) \longrightarrow (\xi^X_{3n}, \Gamma_B(z_0)), \tag{47}$$
$$(\xi^X_{3n}, \Gamma_A(z_0)) \longrightarrow (\xi^X_{4n}, \Gamma_B(z_0)), \tag{48}$$
$$(\xi^X_{4n}, \Gamma_B(z_{-,n})) \longrightarrow (\xi^X_{1n}, \Gamma_A(z_{-,n})). \tag{49}$$

Their continuation contrasts are

$$+\gamma(z_{+,n}), \quad -\gamma(z_0), \quad -\gamma(z_0), \quad +\gamma(z_{-,n}).$$

Theorem 5.6 (Second-order identification in the two-block design)**.** *Suppose that for every n the eight menus* (42)–(49) *are observed, all current-node sets used by these blocks are pairwise disjoint* *across designs, and no other observed comparison uses these current nodes. Assume cancellation and let H be C^2 near z_0.*

Let b_n^R and b_n^X be the sums of the four adjusted exact transfers around the two current cycles. Then

$$b_n^R = H(z_{00,n}) - H(z_{10,n}) - H(z_{01,n}) + H(z_{11,n}), \tag{50}$$

$$b_n^X = H(z_{+,n}) - 2H(z_0) + H(z_{-,n}), \tag{51}$$

and

$$\boxed{\frac{b_n^R}{h_n k_n} \longrightarrow H_{xa}(z_0), \qquad \frac{b_n^X}{h_n^2} \longrightarrow H_{xx}(z_0).} \tag{52}$$

No level $H(z)$ at a sampled point is point identified. Over the reduced-form C^2 class, the observationally null two-jets are exactly

$$\boxed{\left\{(\theta_0, \theta_x, \theta_a, 0, 0, \theta_{aa})^\mathsf{T}\right\}.} \tag{53}$$

Consequently a local two-jet functional is point identified if and only if it is a linear combination of $H_{xx}(z_0)$ and $H_{xa}(z_0)$.

Proof. The current incidences in each set of four menus form a simple directed cycle. Summing the four transfer equations removes every current-node value. The continuation contrasts in (42)–(45) sum to (50); those in (46)–(49) sum to (51). The limits in (52) are the usual mixed rectangle and centered second differences.

Let

$$q(x, a) = \alpha + \beta(x - x_0) + \psi(a),$$

with $\alpha, \beta \in \mathbb{R}$ and $\psi \in C^2$. Its rectangle difference and centered second difference in x are both zero. Replacing H by $H + q$ therefore leaves the continuation sum around every observed current cycle unchanged. Because each current block is a simple cycle and different blocks use disjoint current nodes, current-node values can be adjusted successively along the first three edges; the fourth edge then holds precisely because the cycle sum of q is zero. Hence H and $H + q$ reproduce every primitive transfer observation. In particular, no sampled level $H(z)$ is point identified.

Conversely, every observationally null perturbation has zero normalized rectangle and curvature residuals for every n, and hence has

$$q_{xa}(z_0) = q_{xx}(z_0) = 0.$$

Every two-jet with these two entries zero is realized by

$$q(x, a) = \alpha + \beta(x - x_0) + \gamma(a - a_0) + \frac{\delta}{2}(a - a_0)^2,$$

which is observationally null by the preceding argument. This proves (53). The two finite-difference limits in (52) identify the xx and xa directions, and the null perturbations show that no other two-jet direction is identified. □

Corollary 5.7 (Edge minimality within the two-block design)**.** *If one of the four rectangle-menu positions is absent from every design, then $H_{xa}(z_0)$ is not point identified from the remaining two-block data. If one of the four curvature-menu positions is absent from every design, then $H_{xx}(z_0)$ is not point identified.*

Proof. Deleting any one menu from each rectangle block leaves a tree on its four current nodes, so that block has no nonzero circulation. The remaining curvature blocks cannot identify H_{xa}: the perturbation

$$q(x,a) = (x - x_0)(a - a_0)$$

has zero curvature residuals and changes $q_{xa}(z_0)$. Since the incomplete rectangle blocks are trees, their edge equations can be preserved by current-node adjustments. Thus $H_{xa}(z_0)$ is not identified. Similarly, if one menu is deleted from each curvature block, the rectangle blocks do not see

$$q(x,a) = (x - x_0)^2,$$

while the incomplete curvature blocks are trees; hence $H_{xx}(z_0)$ is not identified. □

Each derivative is carried by one four-edge current cycle, while the continuation values themselves remain unrestricted.

Proposition 5.8 (Additional comparisons)**.** *Suppose an enlarged local comparison support contains the two four-edge blocks of Theorem 5.6. Let $\widetilde{\Lambda}_n$ denote the A–B contrast space generated by all observed current circulations in the enlarged design. Suppose that every $\lambda \in \widetilde{\Lambda}_n$ satisfies*

$$\sum_{i:a_{in}=\bar{a}} \lambda_i = 0 \quad \textit{for every sampled state } \bar{a}, \tag{54}$$

$$\sum_i \lambda_i (x_{in} - x_0) = 0. \tag{55}$$

Then, over the reduced-form C^2 class, the identified two-jet functionals remain exactly

$$\boxed{\operatorname{span}\{\partial_{xx}, \partial_{xa}\}.} \tag{56}$$

In particular, the two response-relevant derivatives remain point identified and no level $H(z)$ is point identified. The additional comparisons may share current nodes with the original blocks.

Proof. The original two blocks remain observed, so every observationally equivalent perturbation has

$$q_{xx}(z_0) = q_{xa}(z_0) = 0.$$

Conversely, for

$$q(x,a) = \alpha + \beta(x - x_0) + \psi(a), \qquad \psi \in C^2,$$

conditions (54)–(55) give

$$\sum_i \lambda_i q(z_{in}) = 0 \qquad \forall \lambda \in \widetilde{\Lambda}_n.$$

Thus these perturbations remain observationally null in the enlarged circulation system. Their two-jets exhaust all vectors with zero xx and xa entries. Together with the original rectangle and centered second differences, this proves (56). A constant perturbation shows that no pointwise level is identified. □

5.5 Higher-order local information

The same argument is not specific to two-jets. Let $m \geq 1$ and let

$$\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}_0^2, \qquad |\alpha| = \alpha_1 + \alpha_2, \qquad \alpha! = \alpha_1!\alpha_2!.$$

For

$$\delta z_{in} = (\delta x_{in}, \delta a_{in}) = z_{in} - z_0,$$

write

$$(\delta z_{in})^\alpha = \delta x_{in}^{\alpha_1} \delta a_{in}^{\alpha_2}.$$

Let $t = (t_\alpha)_{|\alpha| \leq m}$ and define

$$\mathcal{L}_t^{(m)} f(z_0) := \sum_{|\alpha| \leq m} t_\alpha \, \partial^\alpha f(z_0). \tag{57}$$

Theorem 5.9 (Higher-order finite differences)**.** *Suppose the support radius of the local designs tends to zero. Let $\lambda_n \in \Lambda_n$ and let $s_n \neq 0$ satisfy, for every multi-index $|\alpha| \leq m$,*

$$\frac{1}{s_n} \sum_i \lambda_{in} \frac{(\delta z_{in})^\alpha}{\alpha!} \longrightarrow t_\alpha, \tag{58}$$

together with

$$\frac{1}{|s_n|} \sum_i |\lambda_{in}| \|\delta z_{in}\|^m = O(1). \tag{59}$$

Then for every $f \in C^m$ near z_0,

$$\boxed{\frac{1}{s_n} \sum_i \lambda_{in} f(z_{in}) \longrightarrow \mathcal{L}_t^{(m)} f(z_0).} \tag{60}$$

Under cancellation, the same limit is identified from the corresponding normalized circulation sums.

Proof. Taylor's formula at z_0 gives, uniformly over the shrinking support,

$$f(z_{in}) = \sum_{|\alpha| \leq m} \frac{\partial^\alpha f(z_0)}{\alpha!} (\delta z_{in})^\alpha + R_{in}, \qquad R_{in} = o(\|\delta z_{in}\|^m).$$

After multiplication by λ_{in}/s_n and summation, the polynomial part converges by (58). The remainder tends to zero by (59). For $f = H$, the left side equals the normalized observed circulation transfer by (29). □

Let Π_{m-1} be the polynomials in (x, a) of total degree at most $m - 1$.

Theorem 5.10 (Higher-order identification without lower-order recovery)**.** *Suppose every observed A–B circulation contrast annihilates lower-order polynomials:*

$$\sum_i \lambda_i p(z_{in}) = 0 \qquad \textit{for every } n, \ \lambda \in \Lambda_n, \ p \in \Pi_{m-1}. \tag{61}$$

Suppose also that Theorem 5.9 holds for a coefficient vector t with

$$t_\alpha = 0 \qquad \textit{whenever } |\alpha| < m.$$

Then the homogeneous mth-order functional

$$\boxed{\sum_{|\alpha|=m} t_\alpha \, \partial^\alpha H(z_0)} \tag{62}$$

is point identified, while no nonzero local functional involving only derivatives of orders strictly below m is point identified over the reduced-form C^m class.

More generally, if such finite-difference weight sequences span a subspace $\mathcal{S}_m$ of the homogeneous mth-order derivative space, every element of $\mathcal{S}_m$ is identified although the entire $(m-1)$-jet remains unidentified.

Proof. Identification of (62) follows from Theorem 5.9. By (61), every $p \in \Pi_{m-1}$ is observationally null. Given any nonzero linear functional of the $(m-1)$-jet, choose its Taylor polynomial $p \in \Pi_{m-1}$ with that functional nonzero at z_0. Adding arbitrary multiples of p leaves every cycle equation unchanged and moves the target through all real values. Hence no nonzero lower-order jet functional is point identified. The same argument applies simultaneously to every such sequence spanning $\mathcal{S}_m$. □

The second-order design in Theorem 5.6 is the case $m = 2$. Its circulation weights annihilate affine functions, while its second-order moments span

$$\operatorname{span}\{\partial_{xx}, \partial_{xa}\}.$$

The additional null direction in ∂_{aa} is determined by the support geometry of that particular design.

Corollary 5.11 (Counterfactual comparative static without level recovery)**.** *Under the conditions of Theorem 5.6, suppose the compensated observations in A identify $W_{A,xa}$ and $W_{A,xx}$, the current-side derivatives d_{xa} and d_{xx} are known, and the target baseline is an interior strict local optimum. Then $D_a g_B(a_0)$ is point identified even though $H(z)$ is not point identified at any nearby design point.*

Proof. The theorem identifies $\eta_{xa} = H_{xa}(z_0)$ and $\eta_{xx} = H_{xx}(z_0)$. Substitution into the transport identities and the target first-order condition gives (69). □

6 Counterfactual comparative statics

Let A be the observed anchor environment and B the target environment. Write

$$W_J(x,a) = u_J(x,a) + V(\Gamma_J(x,a)), \qquad J \in \{A, B\}, \tag{63}$$

where the current-side difference $u_B - u_A$ is known. Set

$$d_{xa} := (u_B - u_A)_{xa}, \tag{64}$$

$$d_{xx} := (u_B - u_A)_{xx}. \tag{65}$$

Since $H = V \circ \Gamma_A - V \circ \Gamma_B$, the corrections identified in Section 5 satisfy

$$\boxed{W_{B,xa} = W_{A,xa} + d_{xa} - \eta_{xa},} \tag{66}$$

$$\boxed{W_{B,xx} = W_{A,xx} + d_{xx} - \eta_{xx}.} \tag{67}$$

Let $C_B(x,a)$ be the monetary resource cost in the target environment. The baseline target choice $x = g_B(a_0)$ is observed. Assume it is an interior strict local optimum and

$$C_{B,xx} - W_{B,xx} > 0 \tag{68}$$

at the point considered.

Theorem 6.1 (Counterfactual comparative static)**.** *At $(g_B(a_0), a_0)$, suppose the compensated observations in A identify $W_{A,xa}$ and $W_{A,xx}$, the current-side derivatives d_{xa} and d_{xx} are known, and the local comparison designs contain finite-difference weights satisfying* (32)*–*(33) *for ∂_{xa} and ∂_{xx}. Then $D_a g_B(a_0)$ is point identified and*

$$\boxed{D_a g_B(a_0) = \frac{W_{A,xa} + d_{xa} - \eta_{xa} - C_{B,xa}}{C_{B,xx} - W_{A,xx} - d_{xx} + \eta_{xx}}.} \tag{69}$$

Proof. The target first-order condition is

$$W_{B,x}(g_B(a), a) = C_{B,x}(g_B(a), a).$$

Differentiation along the regular local branch gives

$$(C_{B,xx} - W_{B,xx}) D_a g_B = W_{B,xa} - C_{B,xa}.$$

Use (66)–(67) and Theorem 5.1 to identify η_{xa} and η_{xx}. □

When nearby compensated comparisons are observed in A, (12) gives

$$W_{A,xa} = \pi_{A,a}, \qquad W_{A,xx} = \pi_{A,x}, \tag{70}$$

and therefore

$$\boxed{D_a g_B(a_0) = \frac{\pi_{A,a} + d_{xa} - \eta_{xa} - C_{B,xa}}{C_{B,xx} - \pi_{A,x} - d_{xx} + \eta_{xx}}.} \tag{71}$$

When nearby compensated comparisons are also observed in B, (18) applies directly.

6.1 Higher-order comparative statics

Let

$$F_B(x,a) = W_{B,x}(x,a) - C_{B,x}(x,a), \qquad F_B(g_B(a), a) = 0.$$

Repeated implicit differentiation is standard. Once $D_a g_B(a_0), \ldots, D_a^{r-1} g_B(a_0)$ are known, the only new order-$(r+1)$ continuation term entering $D_a^r g_B(a_0)$ is the directional functional

$$\boxed{(\partial_a + D_a g_B(a_0)\partial_x)^r H_x(z_0), \qquad z_0 = (g_B(a_0), a_0),} \tag{72}$$

with the coefficient $D_a g_B(a_0)$ held fixed in the differential operator. Hence Theorem 5.9 identifies the rth policy derivative whenever the circulation moments generate (72), together with the known reference, current-side, and cost derivatives. The full order-$(r+1)$ continuation jet need not be recovered. For example, the new third-order continuation term needed for $D_a^2 g_B(a_0)$ is

$$H_{xaa} + 2(D_a g_B) H_{xxa} + (D_a g_B)^2 H_{xxx}.$$

Thus the order separation in Section 5.5 passes directly to higher-order comparative statics.

7 An extension to several environments

Let $\mathcal{J}$ be a finite set of environments and write

$$\Psi_J = V \circ \Gamma_J.$$

For any linear local functional $\mathcal{L}$ already identified from comparison data, define the observed pairwise correction

$$\eta_{JK}^{\mathcal{L}} = \mathcal{L}(\Psi_J - \Psi_K).$$

These are ordinary potential differences on a graph whose nodes are environments.

Proposition 7.1 (Path independence)**.** *Let* B *be the node-edge incidence matrix of the environment graph and let* $\eta^{\mathcal{L}}$ *collect its oriented edge corrections. The following are equivalent:*

(i) there are node potentials θ_J *such that* $\eta_{JK}^{\mathcal{L}} = \theta_J - \theta_K$ *on every observed edge;*

(ii) $c^{\mathsf{T}}\eta^{\mathcal{L}} = 0$ *for every* $c \in \ker \mathsf{B}$*;*

(iii) path sums between two connected environments are path independent.

Under the common continuation representation, one may take $\theta_J = \mathcal{L}\Psi_J$*. Hence a spanning tree is sufficient to recover every reference-to-target correction in a connected component, while each independent non-tree edge supplies one overidentifying cycle restriction.*

Proof. The first condition is $\eta^{\mathcal{L}} = \mathsf{B}^{\mathsf{T}}\theta$. Since $\mathrm{col}(\mathsf{B}^{\mathsf{T}}) = (\ker \mathsf{B})^{\perp}$, it is equivalent to the second. The path statement follows by telescoping. □

This extension introduces no second identification mechanism. The current-comparison graph identifies the edge corrections; the environment graph only adds already identified potential differences. If a target lies outside the reference environment's connected component, its reference-to-target correction is not determined by these edge differences alone.

8 A dynamic consequence

Let the inherited state be $s \in \mathbb{R}^d$, the current action $x \in \mathbb{R}^m$, and the known technological law of motion be

$$s^+ = T(s, x).$$

If the comparison analysis identifies a differentiable policy $x = g(s)$, the induced closed-loop map is

$$\Phi(s) := T(s, g(s)). \tag{73}$$

By the chain rule,

$$\boxed{D\Phi(s) = D_sT(s, g(s)) + D_xT(s, g(s))Dg(s).} \tag{74}$$

Thus identification of the policy derivative identifies the Jacobian of the state transition. Higher transition derivatives follow from the ordinary multivariate chain rule whenever the corresponding policy jet is identified.

From this point the implications are standard for discrete-time dynamical systems (Elaydi, 2005; Kuznetsov, 2004). At a hyperbolic fixed point, the multipliers of (74) give the usual local stability and saddle implications. At a critical multiplier, higher transition derivatives can matter. For a scalar fixed point normalized to zero with $\Phi'(0) = -1$, the conventional cubic flip coefficient is

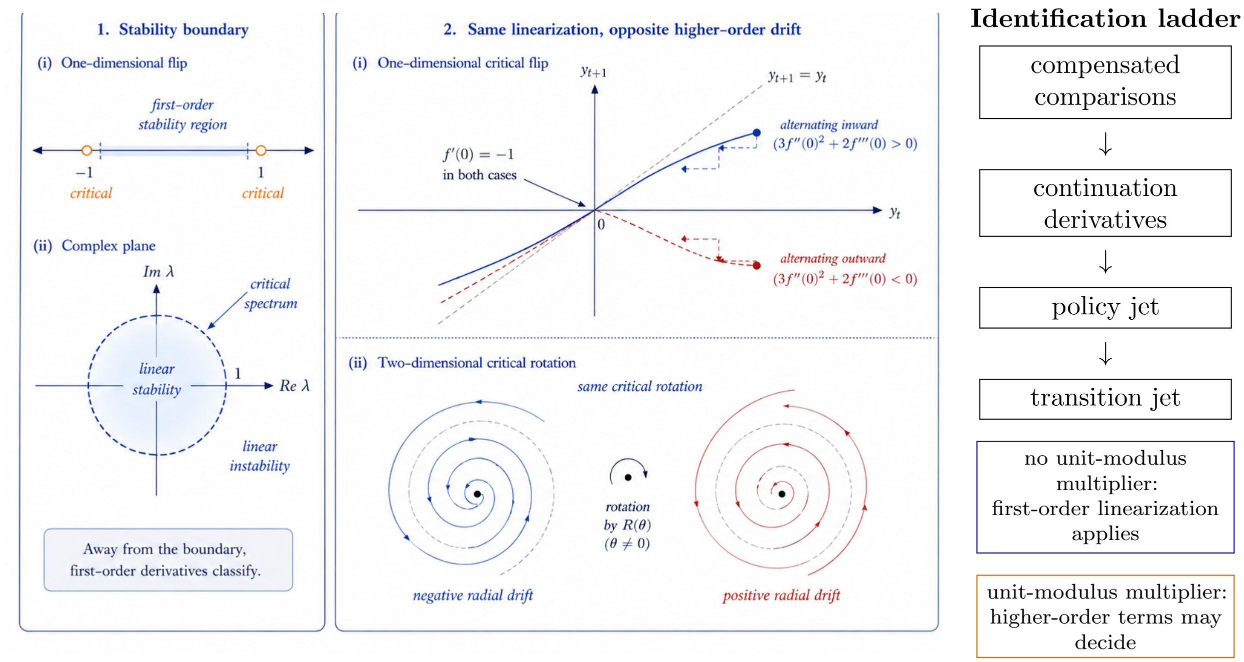
$$\boxed{c_{\text{flip}} = \frac{1}{4}\Phi''(0)^2 + \frac{1}{6}\Phi'''(0).} \tag{75}$$

Equivalently,

$$\Phi^2(y) = y - 2c_{\text{flip}}y^3 + o(|y|^3).$$

Hence the comparison data identify this standard coefficient whenever they identify the third-order transition jet. A nonreal pair of unit-modulus multipliers is the standard Neimark–Sacker critical configuration; again, the identification question concerns the transition derivatives and normal-form coefficient, not the dynamical classification itself. Figure 4 summarizes this distinction.

Figure 4. Critical dynamics beyond the Jacobian



Under the stationary discounted specialization, standard Bellman verification and value iteration may be applied to the recovered policy (Bellman, 1957; Stokey, Lucas, and Prescott, 1989). Recursion is therefore a representation and computational device after identification, not an identifying restriction.

9 Domain and interpretation

The monetary primitive is a transferable numeraire: exact compensatory transfers are comparable on one additive scale. Cancellation determines whether the observed transfers admit the additive representation (21). Its potentials are finite-data representation coordinates, not values assumed from the structural decomposition in Section 3.

The target environment and the observed comparison family are distinct. The target continuation map is defined on its local choice set even when some compensated comparisons are absent from the data. Throughout the local results, "identified" refers to the reduced-form smooth continuation class satisfying the observed circulation equations unless a structural qualification is stated. Restrictions generated by a fixed common V and fixed continuation maps (Γ_A, Γ_B) may further shrink the structural identified set.

The two-stage insurance example isolates the substantive support requirement. A current insurance consequence can be held fixed while the state-contingent renewal clause is varied in an elicitation menu, with a premium transfer restoring indifference. Such contract variation creates a non-singleton continuation fiber. If instead the future consequence is technologically determined by the current consequence, Corollary 5.5 gives zero continuation image. The theorem therefore distinguishes assignable continuation contracts from technologically fixed continuations rather than treating recombination as a property of every dynamic choice problem.

The smooth results use shrinking circulation relations, Taylor expansion, and limits. Higher-order comparative statics follow by repeated differentiation of the first-order condition. The environment graph, when used, contains potential differences already identified from the current-comparison graph. The dynamic section composes the identified policy with a technological law of motion; all subsequent stability and recursive arguments are standard consequences of the recovered policy and transition derivatives.

10 Conclusion

An exact compensated comparison places a finite current change on a common monetary scale. Repetition across nearby states gives the direct comparative static when the target environment itself is observed locally. When it is not, circulations in the observed comparison graph eliminate unrestricted current-node values and leave continuation differences.

The finite information content of a comparison design is the continuation image $Q(\ker D)$. Its dimension is a rank difference, and a new comparison adds a direction only when it is redundant on the current side but not on the joint current–continuation side. This turns the graphical intuition of a closed current projection and a non-closing continuation lift into an edge-by-edge information statement.

Locally, shrinking circulation weights can identify derivatives without identifying levels. In the two-block design, H_{xa} and H_{xx} are point identified while every sampled value $H(z)$ remains unidentified; removing one edge from the relevant block destroys the corresponding derivative. The same argument extends by derivative order: lower-order polynomial terms may be observationally null while a higher-order moment survives.

These identified continuation functionals enter the target first-order condition and determine the counterfactual comparative static. The support condition has a direct contractual interpretation: the current consequence must sometimes admit more than one observed continuation clause. Potential differences across environments and standard state-space dynamics can then be applied to the recovered objects without supplying additional identifying restrictions.

The present account may close exactly; what remains is the price of what follows.

A Proofs of the finite graph theorems

Lemma A.1 (Rational null-space basis)**.** *If a matrix has rational entries, its null space has a basis with rational coordinates.*

Proof. Row reduction over $\mathbb{Q}$ expresses pivot variables as rational linear combinations of free variables. Setting the free variables successively equal to standard basis vectors gives a rational basis. □

Lemma A.2 (Clearing denominators)**.** *Let M be an integer matrix and ℓ an integer vector. If ℓ belongs to the real column space of M, then there exist $k \in \mathbb{N}_+$ and $z \in \mathbb{Z}^{\dim(domain\ M)}$ such that*

$$k\ell = Mz.$$

Proof. The system $Mz = \ell$ has a real solution. Gaussian elimination over $\mathbb{Q}$ gives a rational solution; multiply by a common denominator. □

Proof of Theorem 4.2. Let

$$M = \begin{pmatrix} D \\ Q \end{pmatrix}.$$

If (21) holds and $Mz = 0$, then $z^\mathsf{T}\tau = 0$, so cancellation is necessary.

Conversely, M has integer entries. By Theorem A.1, every real vector in $\ker M$ is a real linear combination of rational null vectors; clearing denominators reduces these to integer null vectors. Hence cancellation implies

$$Mz = 0 \Longrightarrow z^\mathsf{T}\tau = 0 \qquad (z \in \mathbb{R}^E).$$

Thus $\tau \perp \ker M$, so τ lies in the row space of M, equivalently in the column space of M^T. Therefore there are u and v with

$$\tau = M^\mathsf{T}(u, v) = D^\mathsf{T}u + Q^\mathsf{T}v.$$

□

Proof of Theorem 4.4. Let $w = (u, v)$ and $M = (D^\mathsf{T}, Q^\mathsf{T})$ so that the observation equation is $Mw = \tau$. The target functional is

$$\ell^\mathsf{T}w, \qquad \ell = (0, c).$$

It is constant on all solutions of $Mw = \tau$ if and only if ℓ is orthogonal to $\ker M$, equivalently if and only if ℓ belongs to the row space of M. Hence there is a real edge vector z such that

$$Dz = 0, \qquad Qz = c.$$

Because D, Q, c are integer, Theorem A.2 gives the finite-replication form (23). Multiplying (21) by z^T gives (24).

If ℓ is not in the row space of M, there is a perturbation $h \in \ker M$ with $\ell^\mathsf{T}h \neq 0$. Starting from any solution, $w + th$ remains observationally equivalent for all $t \in \mathbb{R}$, while the target functional ranges over all of $\mathbb{R}$. □

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